

DRY-RUN GUIDE | 17_valid_parentheses.py
Source: Python\dsa_solutions\17_valid_parentheses.py

CORE BEHAVIOR

Uses a stack to match closing brackets with the most recent unmatched opening bracket.

REPRESENTATIVE INPUT

'([])'

STEP TRACE

Push (, push [,] pops [,] pops (; stack ends empty.

EXPECTED OUTPUT / RETURN

True

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\dsa_solutions\17_valid_parentheses.py; line numbers refer to the original file. No source was modified.

```
001 | import sys
002 |
003 |
004 | def main():
005 |     text = sys.stdin.read().strip()
006 |     pairs = {"(": ")", "[": "]", "{": "}"}
007 |     stack = []
008 |     for character in text:
009 |         if character in "([{":
010 |             stack.append(character)
011 |         elif character not in pairs or not stack or stack.pop() != pairs[character]:
012 |             print("false")
013 |             return
014 |     print("true" if not stack else "false")
015 |
016 |
017 | if __name__ == "__main__":
018 |     main()
019 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.